

Erdős–Turán, Berry–Esseen, and Edgeworth Universality after One Shuffle

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Edgeworth and optimality working branch

Abstract

We prove functional Erdős–Turán universality, a sharp Berry–Esseen theorem, and a universal first Edgeworth expansion for several nonexchangeable random-permutation models generated by a single shuffle. The common input is an exponentially Ewens-stable cycle index: under geometric randomization of the size, its logarithm differs from the uniform-permutation cycle index by coefficients that decay exponentially in the cycle length. For every such triangular family, if $C_{n,r}$ denotes the number of r -cycles, then

$$\left(\frac{\sum_{r \leq n^t} C_{n,r} - t \log n}{\sqrt{\log n}}, \frac{\log \text{lcm}\{r \leq n^t : C_{n,r} > 0\} - \frac{1}{2}t^2(\log n)^2}{\sqrt{\frac{1}{3}(\log n)^3}} \right)_{0 \leq t \leq 1} \Rightarrow \left(W(t), \sqrt{3} \int_0^t u \, dW(u) \right)_{0 \leq t \leq 1}$$

in $D([0, 1], \mathbb{R}^2)$. In particular, the endpoint correlation between the number of cycles and the logarithm of the order is $\sqrt{3}/2$.

Writing $Y_n = \log \text{ord}(\pi_n)$, $m_n = \mathbb{E}Y_n$, and $\ell_n = \log n$, we prove, uniformly over every subfamily with common stability constants,

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left\{ \frac{Y_n - m_n}{\sqrt{\ell_n^3/3}} \leq x \right\} - \Phi(x) - \frac{\sqrt{3}}{8\sqrt{\ell_n}}(1 - x^2)\varphi(x) \right| \leq C \left(\frac{\log \ell_n}{\ell_n} \right)^{2/3}.$$

Consequently,

$$d_K \left(\frac{Y_n - m_n}{\sqrt{\ell_n^3/3}}, N(0, 1) \right) = \frac{\sqrt{3}}{8\sqrt{2\pi\ell_n}} + O \left(\left(\frac{\log \ell_n}{\ell_n} \right)^{2/3} \right),$$

so the $(\log n)^{-1/2}$ rate is genuinely optimal, with a universal leading constant. We also retain the explicit Barbour–Tavaré form centered at $\frac{1}{2}\ell_n^2 - \ell_n \log \ell_n$.

The proof is a quantitative long-cycle universality argument. Exponential Ewens stability implies that, after deleting cycles of length at most b , the entire remaining cycle vector is within $O(\rho^b)$ in total variation of the corresponding vector for a uniform permutation. A cutoff $b \asymp \log \log n$ transfers the sharp Berry–Esseen upper bound. For the Edgeworth expansion we take $b \asymp \log n$, making the coupling failure polynomially small in n ; the deleted short cycles contribute only $O((\log \log n)^2)$ in mean. Zacharovas’s first Edgeworth expansion for the uniform permutation then transfers with an error $o((\log n)^{-1/2})$.

We verify the criterion for random standardized permutations generated by i.i.d. letters with maximal atom uniformly bounded away from one. This gives all results for arbitrary biased riffle shuffles under the same nonconcentration condition. In particular, a single ordinary

2-riffle shuffle already has the classical functional law, the sharp Berry–Esseen rate, and the universal first Edgeworth correction, although its total-variation distance from uniform tends to one. The same conclusions hold for major-index-biased permutations in the stated uniform parameter range. Using the cycle-index formula of Tripathi, we also prove the results for one asymmetric shelf shuffle, uniformly over an arbitrary number of shelves and over asymmetry parameters bounded away from zero and one.

The arithmetic step is self-contained. We extract a weighted additive-to-least-common-multiple transfer theorem from the von Mangoldt decomposition. Harmonic one-cycle bounds and harmonic two-cycle factorial bounds suffice for the functional law, while the quantitative scalar results follow from the stronger long-cycle coupling.

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1 Introduction

All asymptotic statements are as $n \rightarrow \infty$. We write Φ and φ for the standard normal distribution function and density. For a permutation $\pi \in S_n$, its order is

$$\text{ord}(\pi) = \text{lcm}\{\text{cycle lengths of } \pi\}.$$

Erdős and Turán proved that for a uniform random permutation Π_n ,

$$\frac{\log \text{ord}(\Pi_n) - \frac{1}{2}(\log n)^2}{\sqrt{\frac{1}{3}(\log n)^3}} \Rightarrow N(0, 1) \quad [8].$$

Barbour and Tavaré proved the sharp quantitative form: with $\ell_n = \log n$,

$$d_K\left(\frac{\log \text{ord}(\Pi_n) - \frac{1}{2}\ell_n^2 + \ell_n \log \ell_n}{\sqrt{\ell_n^3/3}}, N(0, 1)\right) = O(\ell_n^{-1/2});$$

see [3] and the explicit formulation recorded in [13]. Zacharovas subsequently proved the first uniform Edgeworth expansion, centered by the exact mean, with correction

$$\frac{\sqrt{3}}{8\sqrt{\log n}}(1 - x^2)\varphi(x)$$

and remainder $O(((\log \log n)/(\log n))^{2/3})$; in particular, the $(\log n)^{-1/2}$ rate is optimal. The subject was placed in the probabilistic theory of logarithmic combinatorial structures; see, among others, [2, 1]. The classical proof reflects two facts. First, cycle counts on logarithmic scales behave like independent Poisson variables of means $1/r$. Second, the logarithm of the product of the cycle lengths can be replaced by the logarithm of their least common multiple.

The present paper shows that this mechanism can be present long before the underlying permutation is globally mixed. Our most concrete result is the following.

Theorem 1.1 (One ordinary 2-riffle shuffle). *Let ρ_n be the permutation obtained from the identity by one ordinary Gilbert–Shannon–Reeds 2-riffle shuffle. Put*

$$\ell_n = \log n, \quad Y_n = \log \text{ord}(\rho_n), \quad m_n = \mathbb{E}Y_n, \quad s_n = \sqrt{\ell_n^3/3}.$$

Then the bivariate functional theorem in Theorem 3.4 holds, and

$$d_K\left(\frac{Y_n - \frac{1}{2}\ell_n^2 + \ell_n \log \ell_n}{s_n}, N(0, 1)\right) \leq \frac{C}{\sqrt{\ell_n}}.$$

Moreover,

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}\left\{\frac{Y_n - m_n}{s_n} \leq x\right\} - \Phi(x) - \frac{\sqrt{3}}{8\sqrt{\ell_n}}(1 - x^2)\varphi(x) \right| \leq C \left(\frac{\log \ell_n}{\ell_n}\right)^{2/3},$$

and therefore

$$d_K\left(\frac{Y_n - m_n}{s_n}, N(0, 1)\right) = \frac{\sqrt{3}}{8\sqrt{2\pi\ell_n}} + O\left(\left(\frac{\log \ell_n}{\ell_n}\right)^{2/3}\right).$$

Thus the normal-approximation rate is optimal. Nevertheless,

$$d_{\text{TV}}(\mathcal{L}(\rho_n), \text{Unif}(S_n)) \rightarrow 1.$$

Thus the order statistic has already reached its uniform-permutation universality class after one minimally nontrivial riffle shuffle, even though the permutation itself remains asymptotically maximally far from uniform.

The proof is organized around a cycle-index criterion that applies to two different mechanisms.

- (i) For standardized permutations, Guerder's primitive-word theorem gives an exact geometric-size decomposition of the cycle types [9]. It yields a cycle index whose non-Ewens coefficients decay exponentially whenever the largest letter probability is bounded away from one.
- (ii) For asymmetric shelf shuffles, Tripathi's exact cycle-index formula has the same analytic form [12]. The unique nondecaying coefficient is precisely the uniform-permutation coefficient; every other coefficient decays exponentially when the asymmetry parameter stays away from zero and one.

The common probabilistic theorem is stronger than an endpoint central limit theorem. Put

$$K_n(t) = \sum_{r \leq n^t} C_{n,r}, \quad \mathcal{O}_n(t) = \log \text{lcm}\{r \leq n^t : C_{n,r} > 0\}, \quad 0 \leq t \leq 1.$$

We prove

$$\left(\frac{K_n(t) - t \log n}{\sqrt{\log n}}, \frac{\mathcal{O}_n(t) - \frac{1}{2} t^2 (\log n)^2}{\sqrt{\frac{1}{3} (\log n)^3}} \right) \Rightarrow \left(W(t), \sqrt{3} \int_0^t u \, dW(u) \right).$$

The two coordinates are driven by the same Brownian noise on logarithmic cycle scale. At the endpoint this gives

$$\left(\frac{K_n - \log n}{\sqrt{\log n}}, \frac{\log \text{ord}(\pi_n) - \frac{1}{2} (\log n)^2}{\sqrt{\frac{1}{3} (\log n)^3}} \right) \Rightarrow N \left(0, \begin{pmatrix} 1 & \sqrt{3}/2 \\ \sqrt{3}/2 & 1 \end{pmatrix} \right).$$

A further contribution is an abstract arithmetic transfer theorem. For arbitrary component multiplicities (C_r) , let

$$B = \sum_r C_r \log r, \quad L = \text{lcm}\{r : C_r > 0\}.$$

Writing $D_m = \sum_{m|r} C_r$, the exact identity

$$B - \log L = \sum_m \Lambda(m) (D_m - 1)_+$$

reduces the product-LCM discrepancy to repeated prime-power divisibility. We show that the bounds

$$\mathbb{E} C_r \ll \frac{1}{r}, \quad \mathbb{E} [C_r (C_s - \mathbf{1}_{\{r=s\}})] \ll \frac{1}{rs}$$

already imply

$$\mathbb{E}(B - \log L) = O(\log n \log \log n).$$

The same estimate transfers the entire process, because the truncated product-LCM discrepancy is nondecreasing in the truncation parameter.

Main applications

The standardized-permutation, sharp Berry–Esseen, and Edgeworth theorems allow an arbitrary triangular array of discrete distributions $p^{(n)}$ on finite or countable ordered alphabets, subject only to

$$\sup_n \|p^{(n)}\|_\infty < 1.$$

No limiting alphabet distribution is assumed. It contains:

- one biased riffle shuffle with arbitrary pile probabilities satisfying the same nonconcentration condition;
- one ordinary a_n -riffle shuffle for every sequence $a_n \geq 2$;
- major-index-biased permutations with every fixed parameter $q \in (0, 1)$, and uniformly for $q_n \geq q_0 > 0$.

The shelf-shuffle theorem permits arbitrary shelf numbers $m_n \geq 1$ and parameters

$$\varepsilon \leq p_n \leq 1 - \varepsilon.$$

In particular, it covers a single shelf and a growing number of shelves in one uniform statement.

Relation with earlier work

The cycle theory of riffle shuffles goes back to Diaconis–McGrath–Pitman [7], and generalized riffle distributions were developed by Stanley [11]. Guerdier recently gave a common primitive-word theory for standardized permutations, including riffle and major-index models [9]. The symmetric shelf shuffle was analyzed by Diaconis–Fulman–Holmes [6]; Tripathi extended its cycle index to the asymmetric model [12]. The present results use those exact enumerative inputs but prove the functional Erdős–Turán laws and the arithmetic transfer independently.

The weighted additive-to-LCM argument was motivated by the comparable-rate Luce order problem. No theorem from the Luce papers is used here.

2 Weighted additive-to-LCM transfer

We begin with the model-independent arithmetic step.

Let $(C_r)_{1 \leq r \leq U}$ be arbitrary nonnegative integer-valued random variables. Define

$$B(U) = \sum_{r \leq U} C_r \log r, \quad L(U) = \text{lcm}\{r \leq U : C_r > 0\},$$

with $\text{lcm} \emptyset = 1$, and

$$D_m(U) = \sum_{\substack{r \leq U \\ m|r}} C_r.$$

Theorem 2.1 (Exact weighted transfer). *For every $1 \leq y \leq U$,*

$$B(U) - \log L(U) = \sum_{m \leq U} \Lambda(m) (D_m(U) - 1)_+,$$

and

$$\begin{aligned}\mathbb{E}[B(U) - \log L(U)] &\leq \sum_{m \leq y} \Lambda(m) \mathbb{E} D_m(U) \\ &\quad + \frac{1}{2} \sum_{y < m \leq U} \Lambda(m) \mathbb{E}(D_m(U))_2.\end{aligned}\tag{1}$$

Proof. The identities

$$\log r = \sum_{m|r} \Lambda(m)$$

and

$$\log L(U) = \sum_{m \leq U} \Lambda(m) \mathbf{1}_{\{D_m(U) \geq 1\}}$$

give

$$B(U) - \log L(U) = \sum_{m \leq U} \Lambda(m) \left(D_m(U) - \mathbf{1}_{\{D_m(U) \geq 1\}} \right).$$

The summand in parentheses equals $(D_m(U) - 1)_+$. For every nonnegative integer D ,

$$(D - 1)_+ \leq D, \quad (D - 1)_+ \leq \frac{1}{2}(D)_2.$$

Use the first inequality for $m \leq y$ and the second for $m > y$. □

Corollary 2.2 (Harmonic moment criterion). *Suppose that*

$$\mathbb{E} C_r \leq \frac{A}{r}$$

and

$$\mathbb{E} \left[C_r (C_s - \mathbf{1}_{\{r=s\}}) \right] \leq \frac{B}{rs}$$

for all $r, s \leq U$. Then

$$\mathbb{E}[B(U) - \log L(U)] \leq C \left\{ A \log(eU) \log(eU) + B \frac{\log^2(eU)}{y} \right\}.$$

In particular, with $y = \log^2(eU)$,

$$\mathbb{E}[B(U) - \log L(U)] = O(A \log U \log \log U + B).$$

Proof. The first-moment assumption gives

$$\mathbb{E} D_m(U) \leq \frac{A}{m} \sum_{j \leq U/m} \frac{1}{j} \leq \frac{A}{m} \log \frac{eU}{m}.$$

Also,

$$(D_m(U))_2 = \sum_{\substack{r, s \leq U \\ m|r, m|s}} C_r (C_s - \mathbf{1}_{\{r=s\}}),$$

so

$$\mathbb{E}(D_m(U))_2 \leq \frac{B}{m^2} \left(\sum_{j \leq U/m} \frac{1}{j} \right)^2 \leq \frac{B}{m^2} \log^2 \frac{eU}{m}.$$

Chebyshev's estimate $\sum_{m \leq x} \Lambda(m) = O(x)$ and partial summation give

$$\sum_{m \leq y} \frac{\Lambda(m)}{m} = O(\log(e y)), \quad \sum_{m > y} \frac{\Lambda(m)}{m^2} = O(1/y).$$

Substitution in (1) proves the result. □

For a process version, put

$$B_n(t) = \sum_{r \leq n^t} C_{n,r} \log r, \quad L_n(t) = \text{lcm}\{r \leq n^t : C_{n,r} > 0\}.$$

Lemma 2.3 (Functional transfer). *The function*

$$t \mapsto B_n(t) - \log L_n(t)$$

is nonnegative and nondecreasing. Consequently,

$$\sup_{0 \leq t \leq 1} |B_n(t) - \log L_n(t)| = B_n(1) - \log L_n(1).$$

Thus, if $s_n > 0$ and

$$\frac{B_n(1) - \log L_n(1)}{s_n} \rightarrow 0$$

in probability, then the additive and LCM processes differ by $o_{\mathbb{P}}(1)$ uniformly after division by s_n .

Proof. When the threshold crosses a length r with $C_{n,r} > 0$, the additive statistic increases by $C_{n,r} \log r$, while the logarithm of the least common multiple increases by at most $\log r$. Hence the discrepancy cannot decrease. □

3 Exponentially Ewens-stable cycle indices

We now isolate the probabilistic condition common to standardized permutations and shelf shuffles.

For a permutation π , write $C_r(\pi)$ for its number of r -cycles.

Definition 3.1 (Triangular cycle-index family). For each target size n , let

$$\{\mathbb{P}_{n,m} : m \geq 0\}$$

be probability measures on S_m , with the convention that S_0 contains the empty permutation. Their cycle index is

$$\mathcal{Z}_n(u, \mathbf{x}) = 1 + \sum_{m \geq 1} u^m \mathbb{E}_{n,m} \prod_{r \geq 1} x_r^{C_r}.$$

We study the diagonal random permutation

$$\pi_n \sim \mathbb{P}_{n,n}.$$

Definition 3.2 (Exponential Ewens stability). The triangular family is *exponentially Ewens-stable* if there are real coefficients $b_{n;r,k}$ and constants $A < \infty$, $\rho \in (0, 1)$, independent of n , such that

$$\mathcal{Z}_n(u, \mathbf{x}) = \exp \left\{ \sum_{r \geq 1} \sum_{k \geq 1} \frac{b_{n;r,k}}{r^k} x_r^k u^{rk} \right\} \quad (2)$$

as a formal power series, and

$$|b_{n;r,1} - 1| \leq A\rho^r, \quad (3)$$

$$|b_{n;r,k}| \leq A\rho^{rk}, \quad k \geq 2. \quad (4)$$

The normalization $\mathcal{Z}_n(u, \mathbf{1}) = (1 - u)^{-1}$ is included in the definition, since every coefficient at $\mathbf{x} = \mathbf{1}$ is one.

The coefficient $b_{n;r,1} = 1$ is the uniform-permutation contribution. The remaining terms are allowed to depend arbitrarily on n , but must be exponentially confined to short cycle lengths.

Put

$$\ell_n = \log n, \quad u_{n,r} = \frac{\log r}{\ell_n}.$$

For a bounded function $f : [0, 1] \rightarrow \mathbb{R}$, define the logarithmic cycle field

$$\mathcal{G}_n(f) = \frac{1}{\sqrt{\ell_n}} \sum_{r=1}^n f(u_{n,r}) \left(C_{n,r} - \frac{1}{r} \right).$$

Theorem 3.3 (Gaussian logarithmic cycle field). *Let (π_n) be an exponentially Ewens-stable triangular family. For every fixed collection of bounded Riemann-integrable functions $f_1, \dots, f_d$ on $[0, 1]$,*

$$(\mathcal{G}_n(f_1), \dots, \mathcal{G}_n(f_d)) \Rightarrow \left(\int_0^1 f_1(u) \, dW(u), \dots, \int_0^1 f_d(u) \, dW(u) \right).$$

Equivalently, the limit is centered Gaussian with covariance

$$\text{Cov}(\mathcal{G}(f), \mathcal{G}(g)) = \int_0^1 f(u)g(u) \, du.$$

The main process consequence is the following.

Theorem 3.4 (Functional Erdős–Turán universality). *Let (π_n) be exponentially Ewens-stable, and let $C_{n,r} = C_r(\pi_n)$. Set*

$$K_n(t) = \sum_{r \leq n^t} C_{n,r},$$

and

$$\mathcal{O}_n(t) = \log \text{lcm}\{r \leq n^t : C_{n,r} > 0\}.$$

Then, in $D([0, 1], \mathbb{R}^2)$,

$$\left(\frac{K_n(t) - t\ell_n}{\sqrt{\ell_n}}, \frac{\mathcal{O}_n(t) - \frac{1}{2}t^2\ell_n^2}{\sqrt{\ell_n^3/3}} \right)_{0 \leq t \leq 1} \Rightarrow \left(W(t), \sqrt{3} \int_0^t u \, dW(u) \right)_{0 \leq t \leq 1}. \quad (5)$$

The limiting process is continuous and has covariance kernels

$$\begin{aligned}\text{Cov}(W(s), W(t)) &= \min(s, t), \\ \text{Cov}\left(\sqrt{3} \int_0^s u \, dW(u), \sqrt{3} \int_0^t u \, dW(u)\right) &= \min(s, t)^3, \\ \text{Cov}\left(W(s), \sqrt{3} \int_0^t u \, dW(u)\right) &= \frac{\sqrt{3}}{2} \min(s, t)^2.\end{aligned}$$

Corollary 3.5 (Endpoint joint law). *Under the hypotheses of Theorem 3.4,*

$$\left(\frac{K_n - \log n}{\sqrt{\log n}}, \frac{\log \text{ord}(\pi_n) - \frac{1}{2}(\log n)^2}{\sqrt{\frac{1}{3}(\log n)^3}}\right) \Rightarrow N\left(0, \begin{pmatrix} 1 & \sqrt{3}/2 \\ \sqrt{3}/2 & 1 \end{pmatrix}\right).$$

In particular,

$$\frac{\log \text{ord}(\pi_n) - \frac{1}{2}(\log n)^2}{\sqrt{\frac{1}{3}(\log n)^3}} \Rightarrow N(0, 1).$$

Theorem 3.6 (Sharp Berry–Esseen universality). *Let (π_n) be an exponentially Ewens-stable triangular family with stability constants A and ρ . Put*

$$\ell_n = \log n, \quad a_n = \frac{1}{2}\ell_n^2 - \ell_n \log \ell_n, \quad s_n = \sqrt{\ell_n^3/3}.$$

Then there is $C = C(A, \rho) < \infty$ such that, for all $n \geq 3$,

$$d_K\left(\frac{\log \text{ord}(\pi_n) - a_n}{s_n}, N(0, 1)\right) \leq \frac{C}{\sqrt{\ell_n}}. \quad (6)$$

The estimate is uniform over every triangular subfamily with common stability constants.

Theorem 3.7 (Universal first Edgeworth expansion and optimality). *Let (π_n) be an exponentially Ewens-stable triangular family with common stability constants A and ρ . Put*

$$Y_n = \log \text{ord}(\pi_n), \quad m_n = \mathbb{E}Y_n, \quad \ell_n = \log n, \quad s_n = \sqrt{\ell_n^3/3}.$$

Then there is $C = C(A, \rho) < \infty$ such that, for all sufficiently large n ,

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}\left\{\frac{Y_n - m_n}{s_n} \leq x\right\} - \Phi(x) - \frac{\sqrt{3}}{8\sqrt{\ell_n}}(1 - x^2)\varphi(x) \right| \leq C \left(\frac{\log \ell_n}{\ell_n}\right)^{2/3}. \quad (7)$$

Consequently,

$$d_K\left(\frac{Y_n - m_n}{s_n}, N(0, 1)\right) = \frac{\sqrt{3}}{8\sqrt{2\pi\ell_n}} + O_{A,\rho}\left(\left(\frac{\log \ell_n}{\ell_n}\right)^{2/3}\right). \quad (8)$$

In particular, the $(\log n)^{-1/2}$ convergence rate is optimal. Both statements are uniform over every triangular subfamily with the same stability constants.

The qualitative and quantitative proofs occupy the next sections.

4 Cumulants from the cycle index

Fix cutoffs

$$b_n = \lceil (\log n)^3 \rceil, \quad U_n = \left\lfloor \frac{n}{(\log n)^3} \right\rfloor.$$

For bounded arrays $f_{n,r}^{(j)}$, supported on $b_n < r \leq U_n$, put

$$S_{n,j} = \sum_{b_n < r \leq U_n} f_{n,r}^{(j)} C_{n,r}.$$

Proposition 4.1 (Mixed-cumulant extraction). *Fix $q \geq 1$ and $M < \infty$. There are constants $C_{q,M} < \infty$ and $\rho_{q,M} \in (0, 1)$, depending only on q, M, A, ρ , such that, uniformly over arrays satisfying*

$$|f_{n,r}^{(j)}| \leq M, \quad 1 \leq j \leq q,$$

one has

$$\left| \kappa(S_{n,1}, \dots, S_{n,q}) - \sum_{b_n < r \leq U_n} \frac{\prod_{j=1}^q f_{n,r}^{(j)}}{r} \right| \leq C_{q,M} (1 + \log n)^q \rho_{q,M}^{b_n}. \quad (9)$$

Proof. For $\mathbf{z} = (z_1, \dots, z_q)$, substitute

$$x_r = \exp \left(\sum_{j=1}^q z_j f_{n,r}^{(j)} \right)$$

for $b_n < r \leq U_n$, and $x_r = 1$ otherwise. Since $\mathcal{Z}_n(u, \mathbf{1}) = (1 - u)^{-1}$,

$$\mathbb{E} \exp \left(\sum_{j=1}^q z_j S_{n,j} \right) = [u^n] \frac{1}{1 - u} \exp H_n(u, \mathbf{z}),$$

where

$$H_n(u, \mathbf{z}) = \sum_{b_n < r \leq U_n} \sum_{k \geq 1} \frac{b_{n,r,k}}{rk} \left[\exp \left(k \sum_{j=1}^q z_j f_{n,r}^{(j)} \right) - 1 \right] u^{rk}.$$

For a nonempty set $B \subseteq [q]$, define

$$H_{n,B}(u) = \prod_{j \in B} \frac{\partial}{\partial z_j} H_n(u, \mathbf{z}) \Big|_{\mathbf{z}=0}.$$

Then

$$H_{n,B}(u) = \sum_{b_n < r \leq U_n} \frac{\prod_{j \in B} f_{n,r}^{(j)}}{r} \left[b_{n,r,1} u^r + \sum_{k \geq 2} b_{n,r,k} k^{|B|-1} u^{rk} \right].$$

Write

$$H_{n,B}(u) = A_{n,B}(u) + E_{n,B}(u),$$

where

$$A_{n,B}(u) = \sum_{b_n < r \leq U_n} \frac{\prod_{j \in B} f_{n,r}^{(j)}}{r} u^r.$$

By (3)–(4), the coefficients of $E_{n,B}$ have exponentially summable absolute values, and

$$\sum_{m \geq 0} |[u^m]E_{n,B}(u)| \leq C_{q,M} \rho_1^{b_n} \quad (10)$$

for some $\rho_1 \in (0, 1)$.

The derivative

$$\partial_{z_1} \cdots \partial_{z_q} e^{H_n(u, \mathbf{z})} \Big|_{\mathbf{z}=0}$$

is the sum, over set partitions π of $[q]$, of

$$\prod_{B \in \pi} H_{n,B}(u).$$

The product of the $A_{n,B}$ terms has degree at most $qU_n < n$ for all large n . Therefore

$$[u^n] \frac{1}{1-u} \prod_{B \in \pi} A_{n,B}(u) = \prod_{B \in \pi} A_{n,B}(1).$$

Every remaining product contains at least one $E_{n,B}$, and its total absolute coefficient mass is

$$O_{q,M}((1 + \log n)^q \rho_1^{b_n}) = o(1)$$

by (10). It follows that all joint moments of order at most q differ by $o(1)$ from those of the formal random vector with cumulant generating function $H_n(1, \mathbf{z})$.

Cumulants are universal polynomials in moments. Hence

$$\kappa(S_{n,1}, \dots, S_{n,q}) = \partial_{z_1} \cdots \partial_{z_q} H_n(1, \mathbf{z}) \Big|_{\mathbf{z}=0} + o(1).$$

The derivative on the right is

$$\sum_{b_n < r \leq U_n} \frac{\prod_j f_{n,r}^{(j)}}{r} \left[b_{n,r,1} + \sum_{k \geq 2} b_{n,r,k} k^{q-1} \right].$$

The difference from the main term in (9) is bounded by $C_{q,M} \rho_1^{b_n}$, again using exponential stability. Combining the two bounds and enlarging the constants gives the displayed estimate. $\square$

The same cycle index gives the moment estimates needed for the arithmetic transfer.

Proposition 4.2 (Harmonic one- and two-cycle bounds). *There is a constant $C < \infty$, depending only on A, ρ , such that, for all n and all $r, s \leq n$,*

$$\mathbb{E} C_{n,r} \leq \frac{C}{r},$$

and

$$\mathbb{E} [C_{n,r}(C_{n,s} - \mathbf{1}_{\{r=s\}})] \leq \frac{C}{rs}.$$

Proof. Differentiate (2) with respect to x_r and set $\mathbf{x} = \mathbf{1}$. Since $\mathcal{Z}_n(u, \mathbf{1}) = (1-u)^{-1}$,

$$\mathbb{E} C_{n,r} = [u^n] \frac{1}{1-u} \left(\frac{1}{r} \sum_{k \geq 1} b_{n,r,k} u^{rk} \right).$$

Thus

$$|\mathbb{E}C_{n,r}| \leq \frac{1}{r} \sum_{k \geq 1} |b_{n;r,k}| \leq \frac{C}{r}.$$

For $r \neq s$, two differentiations give

$$\mathbb{E}(C_{n,r}C_{n,s}) = [u^n] \frac{1}{1-u} A_{n,r}(u) A_{n,s}(u),$$

where

$$A_{n,r}(u) = \frac{1}{r} \sum_{k \geq 1} b_{n;r,k} u^{rk}.$$

The absolute coefficient mass of $A_{n,r}$ is at most C/r , proving the claim.

For $r = s$,

$$\mathbb{E}(C_{n,r})_2 = [u^n] \frac{1}{1-u} \left(A_{n,r}(u)^2 + B_{n,r}(u) \right),$$

where

$$B_{n,r}(u) = \frac{1}{r} \sum_{k \geq 2} (k-1) b_{n;r,k} u^{rk}.$$

The first term has absolute coefficient mass $O(r^{-2})$. The second has mass

$$O\left(\frac{\rho^{2r}}{r}\right) = O(r^{-2}),$$

because $\sup_{r \geq 1} r \rho^{2r} < \infty$. □

5 Long-cycle coupling, sharp Berry–Esseen, and Edgeworth transfer

The exponential coefficient condition gives substantially more than fixed cumulant control: after a growing short-cycle cutoff, the complete remaining cycle vector is close in total variation to that of a uniform permutation. This permits direct transfer of the sharp classical rate and of the first classical Edgeworth term.

Let Π_n be uniform on S_n , and write $C_{n,r}^0 = C_r(\Pi_n)$. For an integer $b \geq 1$, set

$$\mathbf{C}_n^{>b} = (C_{n,r})_{r>b}, \quad \mathbf{C}_n^{0,>b} = (C_{n,r}^0)_{r>b}.$$

Proposition 5.1 (Exponential long-cycle total-variation universality). *For every exponentially Ewens-stable triangular family, there is $C = C(A, \rho) < \infty$ such that, uniformly in n and $b \geq 1$,*

$$d_{\text{TV}}\left(\mathcal{L}(\mathbf{C}_n^{>b}), \mathcal{L}(\mathbf{C}_n^{0,>b})\right) \leq C \rho^b. \quad (11)$$

Proof. Let

$$d_{n;r,k} = b_{n;r,k} - \mathbf{1}_{\{k=1\}}.$$

The uniform-permutation cycle index is

$$\mathcal{Z}_0(u, \mathbf{x}) = \exp \left\{ \sum_{r \geq 1} \frac{x_r u^r}{r} \right\}.$$

Since both cycle indices equal $(1 - u)^{-1}$ at $\mathbf{x} = \mathbf{1}$,

$$\exp \left\{ \sum_{r,k \geq 1} \frac{d_{n;r,k}}{rk} u^{rk} \right\} = 1.$$

The exponent has zero constant term, so it vanishes as a formal power series. After setting $x_r = 1$ for $r \leq b$, we therefore have the exact factorization

$$\mathcal{Z}_n^{>b}(u, \mathbf{x}) = \mathcal{Z}_0^{>b}(u, \mathbf{x}) \exp R_{n,b}(u, \mathbf{x}), \quad (12)$$

where

$$R_{n,b}(u, \mathbf{x}) = \sum_{r > b} \sum_{k \geq 1} \frac{d_{n;r,k}}{rk} (x_r^k - 1) u^{rk}.$$

For an absolutely summable multivariate series F , let $\|F\|_1$ denote its total absolute coefficient mass. Exponential stability gives

$$\begin{aligned} \|R_{n,b}\|_1 &\leq 2A \sum_{r > b} \frac{\rho^r}{r} + 2A \sum_{r > b} \sum_{k \geq 2} \frac{\rho^{rk}}{rk} \\ &\leq C\rho^b. \end{aligned}$$

The coefficient ℓ^1 norm is submultiplicative, and hence

$$\|e^{R_{n,b}} - 1\|_1 \leq e^{\|R_{n,b}\|_1} - 1 \leq C\rho^b. \quad (13)$$

For each $m \geq 0$, the coefficient polynomial $[u^m] \mathcal{Z}_0^{>b}$ has nonnegative coefficients whose sum is

$$[u^m] \mathcal{Z}_0^{>b}(u, \mathbf{1}) = [u^m](1 - u)^{-1} = 1.$$

Taking the coefficient of u^n in (12), using convolution and (13), gives

$$\sum_{\mathbf{c}} \left| \mathbb{P}\{\mathbf{C}_n^{>b} = \mathbf{c}\} - \mathbb{P}\{\mathbf{C}_n^{0,>b} = \mathbf{c}\} \right| \leq C\rho^b.$$

Division by two proves (11). □

We import the sharp classical endpoint theorem in the following form.

Input theorem 5.2 (Barbour–Tavaré sharp rate [3, 13]). *Let*

$$Y_n^0 = \log \text{ord}(\Pi_n), \quad \ell_n = \log n, \quad a_n = \frac{1}{2} \ell_n^2 - \ell_n \log \ell_n, \quad s_n = \sqrt{\ell_n^3/3}.$$

Then

$$\text{d}_K \left(\frac{Y_n^0 - a_n}{s_n}, N(0, 1) \right) \leq \frac{C_0}{\sqrt{\ell_n}}. \quad (14)$$

Input theorem 5.3 (Zacharovas first Edgeworth expansion [13]). *Let*

$$Y_n^0 = \log \text{ord}(\Pi_n), \quad m_n^0 = \mathbb{E}Y_n^0, \quad \ell_n = \log n, \quad s_n = \sqrt{\ell_n^3/3}.$$

Then, for all sufficiently large n ,

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left\{ \frac{Y_n^0 - m_n^0}{s_n} \leq x \right\} - \Phi(x) - \frac{\sqrt{3}}{8\sqrt{\ell_n}} (1 - x^2) \varphi(x) \right| \leq C_1 \left(\frac{\log \ell_n}{\ell_n} \right)^{2/3}. \quad (15)$$

The following elementary lemmas record the quantitative coupling transfer.

Lemma 5.4 (Kolmogorov transfer under a partial coupling). *Let X, X_0 be coupled real random variables. Suppose there are an event G , a nonnegative random variable Δ , and numbers $\eta, d \geq 0$ such that*

$$\mathbb{P}(G^c) \leq \eta, \quad |X - X_0| \leq \Delta \quad \text{on } G, \quad \mathbb{E}\Delta \leq d.$$

Then, for every $a \in \mathbb{R}$, $s > 0$, and $h > 0$,

$$d_K\left(\frac{X - a}{s}, N(0, 1)\right) \leq d_K\left(\frac{X_0 - a}{s}, N(0, 1)\right) + \eta + C\frac{h}{s} + \frac{d}{h}. \quad (16)$$

Proof. For every $z \in \mathbb{R}$,

$$\mathbb{P}\{X \leq z\} \leq \mathbb{P}\{X_0 \leq z + h\} + \mathbb{P}(G^c) + \mathbb{P}\{\Delta > h\},$$

and the analogous lower bound holds with $z - h$. The standard normal cdf has density bounded by $(2\pi)^{-1/2}$, while Markov's inequality gives $\mathbb{P}\{\Delta > h\} \leq d/h$. Taking the supremum over $z = a + sx$ proves the claim. $\square$

Lemma 5.5 (Distributional transfer under a partial coupling). *Under the hypotheses of Lemma 5.4, for every $h > 0$,*

$$\sup_{z \in \mathbb{R}} |\mathbb{P}\{X \leq z\} - \mathbb{P}\{X_0 \leq z\}| \leq \eta + \frac{d}{h} + \sup_{z \in \mathbb{R}} \mathbb{P}\{z - h < X_0 \leq z + h\}. \quad (17)$$

Proof. The upper bound

$$\mathbb{P}\{X \leq z\} \leq \mathbb{P}\{X_0 \leq z + h\} + \eta + \mathbb{P}\{\Delta > h\}$$

and the corresponding lower bound with $z - h$ imply (17) after Markov's inequality. $\square$

Proof of Theorem 3.6. Put

$$Y_n = \log \text{ord}(\pi_n).$$

Choose

$$b_n = \left\lceil \frac{2 \log \ell_n}{|\log \rho|} \right\rceil$$

when $\ell_n > 1$, and absorb the finitely many remaining n into the constant. By Proposition 5.1, the long-cycle vectors can be coupled so that

$$\mathbb{P}\{\mathbf{C}_n^{>b_n} \neq \mathbf{C}_n^{0,>b_n}\} \leq C\rho^{b_n} \leq C\ell_n^{-2}. \quad (18)$$

Let $Y_n^{>b_n}$ and $Y_n^{0,>b_n}$ be the logarithms of the least common multiples of the cycle lengths exceeding b_n in the two permutations. On the successful coupling event these variables are equal. Moreover,

$$\begin{aligned} 0 \leq Y_n - Y_n^{>b_n} &\leq S_{n,b_n} := \sum_{r \leq b_n} C_{n,r} \log r, \\ 0 \leq Y_n^0 - Y_n^{0,>b_n} &\leq S_{n,b_n}^0 := \sum_{r \leq b_n} C_{n,r}^0 \log r. \end{aligned}$$

Hence, on that event,

$$|Y_n - Y_n^0| \leq S_{n,b_n} + S_{n,b_n}^0.$$

Proposition 4.2, together with $\mathbb{E}C_{n,r}^0 = 1/r$, yields

$$\mathbb{E}(S_{n,b_n} + S_{n,b_n}^0) \leq C \sum_{r \leq b_n} \frac{\log r}{r} \leq C \log^2(eb_n). \quad (19)$$

Apply Lemma 5.4 with $X = Y_n$, $X_0 = Y_n^0$, $h = \ell_n$, and the center and scale from Input Theorem 5.2. Equations (14)–(19) give

$$\begin{aligned} d_K\left(\frac{Y_n - a_n}{s_n}, N(0, 1)\right) &\leq \frac{C}{\sqrt{\ell_n}} + C\ell_n^{-2} + C\frac{\ell_n}{\ell_n^{3/2}} + C\frac{\log^2(eb_n)}{\ell_n} \\ &\leq \frac{C}{\sqrt{\ell_n}}, \end{aligned}$$

which proves (6). $\square$

Proof of Theorem 3.7. Let $Y_n^0 = \log \text{ord}(\Pi_n)$ and $m_n^0 = \mathbb{E}Y_n^0$. Choose a fixed $c > 0$ so large that

$$c|\log \rho| > 4,$$

and put

$$b_n = \lceil c\ell_n \rceil.$$

By Proposition 5.1, the long-cycle vectors can be maximally coupled so that, for an event G_n ,

$$\mathbb{P}(G_n^c) \leq C\rho^{b_n} = O(n^{-4}). \quad (20)$$

After this maximal coupling, sample the short-cycle coordinates according to their respective conditional laws given the long vectors. This extends the coupling to the complete cycle vectors without changing either marginal law.

On G_n , the least common multiples of the cycle lengths exceeding b_n are identical. Therefore

$$|Y_n - Y_n^0| \leq \Delta_n := S_{n,b_n} + S_{n,b_n}^0, \quad (21)$$

where

$$S_{n,b} = \sum_{r \leq b} C_{n,r} \log r, \quad S_{n,b}^0 = \sum_{r \leq b} C_{n,r}^0 \log r.$$

Proposition 4.2 and $\mathbb{E}C_{n,r}^0 = 1/r$ give

$$d_n := \mathbb{E}\Delta_n \leq C \sum_{r \leq b_n} \frac{\log r}{r} \leq C \log^2(eb_n) = O((\log \ell_n)^2). \quad (22)$$

We first compare the two laws with the common center m_n^0 . Let

$$\tilde{F}_n(x) = \mathbb{P}\left\{\frac{Y_n - m_n^0}{s_n} \leq x\right\}, \quad F_n^0(x) = \mathbb{P}\left\{\frac{Y_n^0 - m_n^0}{s_n} \leq x\right\}.$$

Set

$$h_n = \ell_n^{3/4}, \quad r_n = \left(\frac{\log \ell_n}{\ell_n}\right)^{2/3}.$$

Lemma 5.5 gives

$$\begin{aligned} \sup_x |\tilde{F}_n(x) - F_n^0(x)| &\leq O(n^{-4}) + \frac{d_n}{h_n} \\ &\quad + \sup_z \mathbb{P}\{z - h_n < Y_n^0 \leq z + h_n\}. \end{aligned} \quad (23)$$

Define the Edgeworth approximant

$$E_n(x) = \Phi(x) + \frac{\sqrt{3}}{8\sqrt{\ell_n}}(1 - x^2)\varphi(x).$$

The functions E_n are uniformly Lipschitz: indeed,

$$E'_n(x) = \varphi(x) + \frac{\sqrt{3}}{8\sqrt{\ell_n}}(x^3 - 3x)\varphi(x),$$

and the polynomial times Gaussian factor is bounded. Input Theorem 5.3 therefore yields

$$\sup_z \mathbb{P}\{z - h_n < Y_n^0 \leq z + h_n\} \leq C \frac{h_n}{s_n} + Cr_n. \quad (24)$$

Since

$$\frac{d_n}{h_n} = O\left(\frac{(\log \ell_n)^2}{\ell_n^{3/4}}\right) = o(r_n), \quad \frac{h_n}{s_n} = O(\ell_n^{-3/4}) = o(r_n),$$

equations (23) and (24) give

$$\sup_x |\tilde{F}_n(x) - F_n^0(x)| = O_{A,\rho}(r_n). \quad (25)$$

It remains to replace the uniform mean by the actual mean. For every permutation of n letters,

$$0 \leq \log \text{ord}(\pi) \leq \sum_{C \in \text{Cyc}(\pi)} \log |C| \leq \sum_{C \in \text{Cyc}(\pi)} |C| = n.$$

Hence, using (20), (21), and (22),

$$|m_n - m_n^0| \leq d_n + 2n\mathbb{P}(G_n^c) = O((\log \ell_n)^2). \quad (26)$$

Put

$$\gamma_n = \frac{m_n - m_n^0}{s_n}.$$

Then $|\gamma_n| = O((\log \ell_n)^2 / \ell_n^{3/2}) = o(r_n)$. Since

$$\mathbb{P}\left\{\frac{Y_n - m_n}{s_n} \leq x\right\} = \tilde{F}_n(x + \gamma_n),$$

Input Theorem 5.3, (25), and the uniform Lipschitz bound for E_n prove (7).

Finally, elementary calculus gives

$$\sup_{x \in \mathbb{R}} |(1 - x^2)\varphi(x)| = \varphi(0) = \frac{1}{\sqrt{2\pi}}.$$

Taking suprema in (7) proves (8) and hence optimality. $\square$

Remark 5.6 (Why the adjusted center is necessary). The qualitative theorem may be centered at $\frac{1}{2}\ell_n^2$ because $\ell_n \log \ell_n = o(\ell_n^{3/2})$. At Berry–Esseen resolution this is no longer harmless: changing from a_n to $\frac{1}{2}\ell_n^2$ translates the normalized variable by

$$\frac{\ell_n \log \ell_n}{s_n} = \sqrt{3} \frac{\log \ell_n}{\sqrt{\ell_n}},$$

which is larger by a factor $\log \ell_n$ than the sharp $\ell_n^{-1/2}$ error. Thus the Barbour–Tavaré correction is intrinsic to the sharp formulation.

6 The Gaussian cycle field and functional tightness

We first record the logarithmic Riemann sums used throughout.

Lemma 6.1 (Logarithmic Riemann sums). *If f is bounded and Riemann integrable on $[0, 1]$, then*

$$\frac{1}{\log n} \sum_{r=1}^n \frac{f(\log r / \log n)}{r} \longrightarrow \int_0^1 f(u) \, du.$$

For every fixed continuous f , the corresponding truncated sums up to n^t converge uniformly in $t \in [0, 1]$:

$$\sup_{0 \leq t \leq 1} \left| \frac{1}{\log n} \sum_{r \leq n^t} \frac{f(\log r / \log n)}{r} - \int_0^t f(u) \, du \right| \longrightarrow 0.$$

Proof. Put $u_{n,r} = \log r / \log n$. The mesh $\max_r (u_{n,r+1} - u_{n,r})$ is $O((\log n)^{-1})$, and

$$u_{n,r+1} - u_{n,r} = \frac{1}{\log n} \log \left(1 + \frac{1}{r} \right) = \frac{1}{r \log n} + O \left(\frac{1}{r^2 \log n} \right).$$

Thus the displayed sum differs by $o(1)$ from the Riemann sum associated with the partition $(u_{n,r})$. This proves the first assertion for continuous functions. Bounded Riemann-integrable functions follow by upper and lower step-function approximation. For continuous f , the same comparison applied to every initial segment of the partition is uniform in its endpoint, proving the second assertion. $\square$

Proof of Theorem 3.3. First replace the full sums by

$$\mathcal{G}_n^\circ(f) = \frac{1}{\sqrt{\ell_n}} \sum_{b_n < r \leq U_n} f(u_{n,r}) \left(C_{n,r} - \frac{1}{r} \right).$$

Proposition 4.2 gives

$$\mathbb{E} \sum_{r \leq b_n} C_{n,r} = O(\log b_n) = O(\log \log n)$$

and

$$\mathbb{E} \sum_{r > U_n} C_{n,r} = O(\log(n/U_n) + 1) = O(\log \log n).$$

The corresponding deterministic harmonic sums have the same orders. Hence

$$\mathcal{G}_n(f) - \mathcal{G}_n^\circ(f) \longrightarrow 0$$

in L^1 , uniformly for f in a fixed bounded set.

For fixed $f_1, \dots, f_d$, apply Proposition 4.1. The second joint cumulants satisfy

$$\kappa(\mathcal{G}_n^\circ(f_i), \mathcal{G}_n^\circ(f_j)) = \frac{1}{\ell_n} \sum_{b_n < r \leq U_n} \frac{f_i(u_{n,r}) f_j(u_{n,r})}{r} + o(1),$$

which converges to $\int_0^1 f_i f_j$ by Lemma 6.1. A joint cumulant of order $q \geq 3$ is

$$O(\ell_n^{1-q/2}) + o(1),$$

and therefore tends to zero. The first cumulants tend to zero, the second cumulants have the displayed limits, and every fixed higher cumulant tends to zero. Hence all moments of every fixed linear combination converge to the corresponding Gaussian moments. Since the Gaussian law is moment-determinate, the method of moments and Cramér–Wold give the asserted finite-dimensional limit. $\square$

We next prove the process theorem first for the additive cycle field. Define

$$B_n(t) = \sum_{r \leq n^t} (\log r) C_{n,r}.$$

Proposition 6.2 (Bivariate additive functional limit). *In $D([0, 1], \mathbb{R}^2)$,*

$$\left(\frac{K_n(t) - t\ell_n}{\sqrt{\ell_n}}, \frac{B_n(t) - \frac{1}{2}t^2\ell_n^2}{\sqrt{\ell_n^3/3}} \right)_{0 \leq t \leq 1} \Rightarrow \left(W(t), \sqrt{3} \int_0^t u \, dW(u) \right)_{0 \leq t \leq 1}.$$

Proof. We first work on the intermediate window and center by exact expectations. Put

$$\mathbf{X}_n^\circ(t) = \frac{1}{\sqrt{\ell_n}} \sum_{\substack{b_n < r \leq U_n \\ r \leq n^t}} \left(\frac{1}{\sqrt{3}u_{n,r}} \right) (C_{n,r} - \mathbb{E}C_{n,r}).$$

Finite-dimensional convergence follows from Proposition 4.1 and Lemma 6.1. Indeed, the covariance matrix at times s, t converges to

$$\int_0^{\min(s,t)} \begin{pmatrix} 1 \\ \sqrt{3}u \end{pmatrix} \begin{pmatrix} 1 & \sqrt{3}u \end{pmatrix} du.$$

All joint cumulants of order at least three vanish after normalization.

We verify tightness. Consider two adjacent increments over $(s, t]$ and $(t, u]$, and any fixed scalar projections A_n, B_n of these increments before division by $\sqrt{\ell_n}$. Their length supports are disjoint. The moment–cumulant identity gives

$$\mathbb{E}(A_n^2 B_n^2) = \kappa(A_n, A_n, B_n, B_n) + \kappa(A_n, A_n) \kappa(B_n, B_n) + 2\kappa(A_n, B_n)^2.$$

By Proposition 4.1, the main terms in the two mixed cumulants vanish because the supports are disjoint. Moreover, for a deterministic sequence $\eta_n \rightarrow 0$, independent of s, t, u ,

$$\kappa(A_n, A_n) \leq C \sum_{\substack{n^s < r \leq n^t \\ b_n < r \leq U_n}} \frac{1}{r} + \eta_n,$$

and similarly for B_n , while the absolute values of the mixed second and fourth cumulants are at most η_n . Integral comparison gives

$$\sum_{\substack{n^s < r \leq n^t \\ b_n < r \leq U_n}} \frac{1}{r} \leq \ell_n(t-s) + O(b_n^{-1}).$$

After division by ℓ_n^2 and use of $(t-s)(u-t) \leq (u-s)^2/4$, we obtain another deterministic $\tilde{\eta}_n \rightarrow 0$ such that

$$\mathbb{E} \left[\left| \frac{A_n}{\sqrt{\ell_n}} \right|^2 \left| \frac{B_n}{\sqrt{\ell_n}} \right|^2 \right] \leq C(u-s)^2 + \tilde{\eta}_n$$

uniformly in $0 \leq s \leq t \leq u \leq 1$. Lemma B.1, the standard vanishing-error form of Billingsley's two-increment criterion, now yields tightness of each coordinate and hence of the bivariate process.

For completeness, the maximal jump also vanishes. By Proposition 4.2,

$$\mathbb{P} \{ \exists r > b_n : C_{n,r} \geq 2 \} \leq \frac{1}{2} \sum_{r > b_n} \mathbb{E}(C_{n,r})_2 = O(b_n^{-1}).$$

With probability tending to one, every intermediate jump in either normalized coordinate is therefore $O(\ell_n^{-1/2})$. Every subsequential limit is continuous.

It remains to restore the omitted lengths and replace exact means by the classical centers. Proposition 4.2 gives

$$\begin{aligned} \mathbb{E} \sum_{r \leq b_n} C_{n,r} + \mathbb{E} \sum_{r > U_n} C_{n,r} &= O(\log \log n), \\ \mathbb{E} \sum_{r \leq b_n} (\log r) C_{n,r} + \mathbb{E} \sum_{r > U_n} (\log r) C_{n,r} &= O(\ell_n \log \log n). \end{aligned}$$

Since these omitted processes are nondecreasing, the same bounds control their suprema in t . They are negligible after the stated normalizations.

Finally, Proposition 4.1 with $q = 1$ and the bounded weights $\mathbf{1}_{\{r \leq n^t\}}$ and $u_{n,r} \mathbf{1}_{\{r \leq n^t\}}$ gives, uniformly in t ,

$$\begin{aligned} \mathbb{E} K_n(t) &= t \ell_n + O(\log \log n), \\ \mathbb{E} B_n(t) &= \frac{1}{2} t^2 \ell_n^2 + O(\ell_n \log \log n). \end{aligned}$$

This completes the proof. □

7 Proof of the functional order theorem

Proof of Theorem 3.4. Proposition 4.2 and Corollary 2.2, with $U = n$, give

$$\mathbb{E} [B_n(1) - \log \text{ord}(\pi_n)] = O(\ell_n \log \ell_n).$$

Indeed, choose $y = \ell_n^2$. The constants in the harmonic moment bounds are uniform over the triangular family.

By Lemma 2.3,

$$\sup_{0 \leq t \leq 1} |B_n(t) - \mathcal{O}_n(t)| = B_n(1) - \log \text{ord}(\pi_n).$$

Consequently,

$$\sup_{0 \leq t \leq 1} \frac{|B_n(t) - \mathcal{O}_n(t)|}{\ell_n^{3/2}} \longrightarrow 0$$

in L^1 . Apply Proposition 6.2 and Slutsky's theorem in $D([0, 1], \mathbb{R}^2)$. $\square$

8 Random standardized permutations

Let I_n be a finite or countable totally ordered alphabet, and let

$$G_{n,1}, \dots, G_{n,m}$$

be i.i.d. with probability vector

$$p^{(n)} = (p_{n,a})_{a \in I_n}.$$

For a word $g = (g_1, \dots, g_m)$, its standardization is the permutation $\text{std}(g) \in S_m$ obtained by ranking the letters increasingly, breaking ties from left to right. We apply the same letter distribution $p^{(n)}$ at every auxiliary size m , and take the diagonal permutation

$$\sigma_n = \text{std}(G_{n,1}, \dots, G_{n,n}).$$

A word is primitive if it is not a nontrivial power. Two primitive words are conjugate if one is a cyclic rotation of the other. Let $\tilde{\mathcal{Q}}_r(I_n)$ be the set of conjugacy classes of primitive words of length r . For $\alpha \in \tilde{\mathcal{Q}}_r(I_n)$, put

$$p_\alpha = \prod_{j=1}^r p_{n,\alpha_j};$$

this is independent of the representative.

We use the following exact result of Guerdar.

Input theorem 8.1 (Primitive-necklace cycle counts [9]). *For each primitive-necklace class α , let D_α be the number of cycles of σ_n of type α . If $\alpha_1, \dots, \alpha_s$ are distinct classes, of respective lengths $r_1, \dots, r_s$, then for all $\ell_1, \dots, \ell_s \geq 0$,*

$$\mathbb{P}(D_{\alpha_1} \geq \ell_1, \dots, D_{\alpha_s} \geq \ell_s) = \begin{cases} \prod_{j=1}^s p_{\alpha_j}^{\ell_j}, & \sum_j r_j \ell_j \leq n, \\ 0, & \sum_j r_j \ell_j > n. \end{cases}$$

Moreover,

$$C_{n,r} = \sum_{\alpha \in \tilde{\mathcal{Q}}_r(I_n)} D_\alpha.$$

Proposition 8.2 (Standardized-permutation cycle index). *For the auxiliary standardized permutations generated by $p^{(n)}$,*

$$\mathcal{Z}_n(u, \mathbf{x}) = \prod_{r \geq 1} \prod_{\alpha \in \tilde{\mathcal{Q}}_r(I_n)} \frac{1}{1 - p_\alpha x_r u^r}.$$

Equivalently,

$$\mathcal{Z}_n(u, \mathbf{x}) = \exp \left\{ \sum_{r \geq 1} \sum_{k \geq 1} \frac{b_{n;r,k}}{r k} x_r^k u^{rk} \right\},$$

where

$$b_{n;r,k} = r \sum_{\alpha \in \tilde{\mathcal{Q}}_r(I_n)} p_\alpha^k = \sum_{w \in \mathcal{Q}_r(I_n)} p_w^k. \quad (27)$$

Proof. Let N be independent with

$$\mathbb{P}(N = m) = (1 - u)u^m, \quad m \geq 0.$$

Mixing the identity in Input Theorem 8.1 over N gives

$$\mathbb{P}(D_{\alpha_1} \geq \ell_1, \dots, D_{\alpha_s} \geq \ell_s) = \prod_{j=1}^s (p_{\alpha_j} u^{r_j})^{\ell_j}.$$

Thus, under geometric size, the D_α are independent geometric variables with tails

$$\mathbb{P}(D_\alpha \geq \ell) = (p_\alpha u^{|\alpha|})^\ell.$$

Their joint probability generating function is

$$(1 - u)\mathcal{Z}_n(u, \mathbf{x}) = \prod_{\alpha} \frac{1 - p_\alpha u^{|\alpha|}}{1 - p_\alpha x_{|\alpha|} u^{|\alpha|}}.$$

Since the total number of letters equals $\sum_{\alpha} |\alpha| D_\alpha = N$, setting all $x_r = 0$ gives the weighted Witt identity

$$\prod_{\alpha} (1 - p_\alpha u^{|\alpha|}) = 1 - u.$$

The first displayed product follows. Taking logarithms and using that a primitive conjugacy class of length r has exactly r elements gives (27). $\square$

Lemma 8.3 (Möbius formula and exponential stability). *For every $r, k \geq 1$,*

$$b_{n;r,k} = \sum_{d|r} \mu(d) \left(\sum_{a \in I_n} p_{n,a}^{kd} \right)^{r/d}. \quad (28)$$

If

$$\sup_n \|p^{(n)}\|_{\infty} \leq R < 1,$$

then there are $A < \infty$ and $\rho \in (0, 1)$, depending only on R , such that

$$|b_{n;r,1} - 1| \leq A\rho^r, \quad |b_{n;r,k}| \leq A\rho^{rk} \quad (k \geq 2).$$

Proof. Every word has a unique primitive root. Möbius inversion on the divisor lattice gives (28).

Put

$$s_j = \sum_a p_{n,a}^j.$$

Since $\max_a p_{n,a} \leq R$,

$$s_j \leq R^{j-1}.$$

For $k = 1$, the $d = 1$ term in (28) equals one. Every term with $d \geq 2$ has absolute value at most

$$R^{(d-1)r/d} \leq R^{r/2}.$$

For $k \geq 2$, every divisor term is bounded by

$$R^{(kd-1)r/d} = R^{kr-r/d} \leq R^{kr/2}.$$

There are at most r divisors. Choose any

$$\sqrt{R} < \rho < 1.$$

The polynomial divisor factor is absorbed into $A\rho^{rk}$. $\square$

Combining the general theory with Lemma 8.3 proves the main standardized result.

Theorem 8.4 (Standardized-permutation universality). *Let*

$$\sigma_n = \text{std}(G_{n,1}, \dots, G_{n,n}),$$

where the letters are i.i.d. with an arbitrary finite or countable ordered-alphabet distribution $p^{(n)}$. Suppose

$$\sup_n \|p^{(n)}\|_\infty \leq R < 1.$$

Then Theorems 3.3, 3.4, 3.6, and 3.7, and Corollary 3.5, hold uniformly over all such triangular arrays. In particular, the constant in (6) may be chosen to depend only on R .

Remark 8.5 (Microscopic nonuniversality versus logarithmic universality). For a fixed nonuniform alphabet distribution, the number of cycles of a fixed length can converge to a sum of independent geometric variables rather than to a Poisson variable; this is one of the phenomena identified by Guerder [9]. Theorem 8.4 shows that these nonclassical microscopic laws contribute only $O(1)$ to each fixed cumulant. On the expanding logarithmic scale, the universal $1/r$ coefficient dominates.

9 Riffle shuffles and major-index bias

9.1 Biased and ordinary riffle shuffles

The inverse description of a generalized riffle shuffle assigns each card an i.i.d. pile label and then sorts stably by that label; see [4, 11]. Depending on convention, this gives the shuffle permutation or its inverse. Since inversion preserves cycle type, this distinction is immaterial here.

Theorem 9.1 (One biased riffle shuffle). *For each n , perform one generalized riffle shuffle with pile probabilities $p^{(n)} = (p_{n,a})$ on a finite or countable ordered set of piles. If*

$$\sup_n \max_a p_{n,a} < 1,$$

then the Gaussian cycle-field theorem, the bivariate functional Erdős–Turán theorem, the endpoint joint law, the sharp Berry–Esseen estimate (6), and the Edgeworth expansion (7) all hold uniformly under the stated nonconcentration bound.

Corollary 9.2 (One ordinary a_n -riffle shuffle). *Let $a_n \geq 2$ be arbitrary, and perform one ordinary a_n -riffle shuffle. Then Theorem 3.4 holds. Moreover, with $\ell_n = \log n$,*

$$d_K \left(\frac{\log \text{ord}(\rho_n) - \frac{1}{2}\ell_n^2 + \ell_n \log \ell_n}{\sqrt{\ell_n^3/3}}, N(0, 1) \right) \leq \frac{C}{\sqrt{\ell_n}},$$

where C is uniform over all sequences $a_n \geq 2$. With exact centering by $\mathbb{E} \log \text{ord}(\rho_n)$, the Edgeworth expansion (7) and the exact Kolmogorov asymptotic (8) also hold uniformly.

Proof. Use the uniform alphabet distribution on a_n letters. Its largest atom is at most $1/2$. $\square$

Proof of Theorem 1.1. The limit theorem is Corollary 9.2 with $a_n = 2$.

For the total-variation statement, the inverse of a one-step 2-riffle shuffle has at most one descent. Let A_n be this event. Then

$$\mathbb{P}\{\rho_n^{-1} \in A_n\} = 1.$$

A permutation with at most one descent is determined by choosing the set of values in its first increasing run, so there are at most 2^n such permutations. Hence

$$\mathbb{P}\{\Pi_n \in A_n\} \leq \frac{2^n}{n!} \rightarrow 0$$

for a uniform permutation Π_n . Total variation is invariant under inversion, and therefore tends to one. $\square$

9.2 Major-index-biased permutations

For $\pi \in S_n$, write

$$\text{maj}(\pi) = \sum_{j: \pi(j) > \pi(j+1)} j.$$

Let $\text{Maj}_{n,q}$ be the probability measure proportional to $q^{\text{maj}(\pi)}$, where $q \in (0, 1)$.

Lemma 9.3 (Geometric-letter representation). *Let $G_1, \dots, G_n$ be i.i.d. with*

$$\mathbb{P}(G_i = j) = (1 - q)q^j, \quad j \geq 0,$$

and put $\sigma = \text{std}(G_1, \dots, G_n)$. If

$$w_0 = (n, n-1, \dots, 1), \quad \tau = w_0 \sigma^{-1} w_0,$$

then τ has law $\text{Maj}_{n,q}$. Moreover, τ and σ have the same cycle type.

Proof. Fix $\sigma \in S_n$, and write $i_k = \sigma^{-1}(k)$. The event $\text{std}(G) = \sigma$ is equivalent to

$$G_{i_1} \leq \dots \leq G_{i_n},$$

with strict inequality between the k th and $(k+1)$ st terms whenever $i_k > i_{k+1}$. Thus strict inequalities occur at the descent set of σ^{-1} .

For a set $D \subseteq [n-1]$, the elementary partition generating function is

$$\sum_{\substack{0 \leq a_1 \leq \dots \leq a_n \\ a_k < a_{k+1} \text{ for } k \in D}} q^{a_1 + \dots + a_n} = \frac{q^{\sum_{k \in D} (n-k)}}{\prod_{j=1}^n (1 - q^j)}.$$

It follows that

$$\mathbb{P}\{\text{std}(G) = \sigma\} = \frac{q^{\text{comaj}(\sigma^{-1})}}{\sum_{\pi \in S_n} q^{\text{comaj}(\pi^{-1})}},$$

where

$$\text{comaj}(\pi) = \sum_{k \in \text{Des}(\pi)} (n - k).$$

Conjugation by w_0 maps a descent at k to a descent at $n - k$, so

$$\text{maj}(w_0 \sigma^{-1} w_0) = \text{comaj}(\sigma^{-1}).$$

The stated law follows. Finally, inversion preserves cycle lengths and conjugation preserves cycle type. $\square$

Corollary 9.4 (Major-index Erdős–Turán law). *Let $\pi_{n,q}$ have law $\text{Maj}_{n,q}$ for a fixed $q \in (0, 1)$. Then Theorems 3.3, 3.4, 3.6, and 3.7, and Corollary 3.5, hold. The conclusions, including the constant in the sharp Berry–Esseen estimate, are uniform for triangular parameters $q_n \geq q_0 > 0$.*

Proof. The largest atom of the geometric distribution is $1 - q$. Apply Theorem 8.4 and Lemma 9.3. $\square$

10 Asymmetric shelf shuffles

We now turn to a model not generated by ordinary standardization.

An (m, p) -shelf shuffle has m labeled shelves. Cards are dealt sequentially from the bottom, assigned independently and uniformly to a shelf, and placed at the top of that shelf with probability p and at the bottom with probability $q = 1 - p$. The shelf piles are then concatenated in shelf order. Let $\nu_{n,m}^{(p)}$ be the induced law on S_n .

Tripathi proved the following exact cycle-index formula.

Input theorem 10.1 (Tripathi's shelf-shuffle cycle index [12]). *For fixed $m \geq 1$ and $p \in (0, 1)$,*

$$1 + \sum_{n \geq 1} u^n \sum_{\pi \in S_n} \nu_{n,m}^{(p)}(\pi) \prod_{i \geq 1} x_i^{C_i(\pi)} = \exp \left\{ \sum_{i \geq 1} \sum_{r \geq 1} \frac{B_{i,r}^{(m,p)}}{i^r} x_i^r u^{ir} \right\},$$

where

$$B_{i,r}^{(m,p)} = \sum_{d|i} \mu(d) m^{i/d-ir} (p^{dr} - (-q)^{dr})^{i/d}. \quad (29)$$

At $\mathbf{x} = \mathbf{1}$, the generating function equals $(1 - u)^{-1}$.

Lemma 10.2 (Uniform exponential stability for shelf shuffles). *Fix $\varepsilon \in (0, 1/2]$. There are $A_\varepsilon < \infty$ and $\rho_\varepsilon \in (0, 1)$ such that, for all*

$$m \geq 1, \quad \varepsilon \leq p \leq 1 - \varepsilon,$$

the coefficients in (29) satisfy

$$|B_{i,1}^{(m,p)} - 1| \leq A_\varepsilon \rho_\varepsilon^i,$$

and

$$|B_{i,r}^{(m,p)}| \leq A_\varepsilon \rho_\varepsilon^{ir}, \quad r \geq 2.$$

Proof. Put $q = 1 - p$ and

$$\alpha(p) = \sqrt{p^2 + q^2}.$$

Uniformly for $p \in [\varepsilon, 1 - \varepsilon]$,

$$\alpha(p) \leq \alpha_\varepsilon := \sqrt{1 - 2\varepsilon(1 - \varepsilon)} < 1.$$

For every integer $s \geq 2$, monotonicity of ℓ^r norms gives

$$p^s + q^s \leq \alpha(p)^s.$$

When $r = 1$ and $d = 1$, the summand in (29) is

$$(p + q)^i = 1.$$

For $r = 1$ and $d \geq 2$,

$$|m^{i/d-i} (p^d - (-q)^d)^{i/d}| \leq \alpha_\varepsilon^i.$$

For $r \geq 2$ and every $d | i$,

$$|m^{i/d-ir} (p^{dr} - (-q)^{dr})^{i/d}| \leq \alpha_\varepsilon^{ir},$$

because $m^{1-dr} \leq 1$. There are at most i divisors. Choose $\rho_\varepsilon \in (\alpha_\varepsilon, 1)$ and absorb the divisor factor into $A_\varepsilon \rho_\varepsilon^{ir}$. $\square$

Theorem 10.3 (One asymmetric shelf shuffle). *Let*

$$m_n \geq 1, \quad \varepsilon \leq p_n \leq 1 - \varepsilon,$$

and let π_n have law $\nu_{n,m_n}^{(p_n)}$. Then the Gaussian cycle-field theorem, the bivariate functional Erdős–Turán theorem, the endpoint joint law, the sharp Berry–Esseen theorem, and the universal first Edgeworth expansion hold uniformly over these parameters. In particular, with $\ell_n = \log n$,

$$d_K \left(\frac{\log \text{ord}(\pi_n) - \frac{1}{2}\ell_n^2 + \ell_n \log \ell_n}{\sqrt{\ell_n^3/3}}, N(0, 1) \right) \leq \frac{C_\varepsilon}{\sqrt{\ell_n}}.$$

Proof. For each target n , use the cycle-index family with the same parameters (m_n, p_n) at every auxiliary size. Input Theorem 10.1 and Lemma 10.2 verify exponential Ewens stability. Apply Theorems 3.3, 3.4, 3.6, and 3.7. $\square$

Remark 10.4. The theorem is uniform in the number of shelves. It includes a single shelf, a fixed number of shelves, and an arbitrarily growing number of shelves. The parameter restriction excludes only degeneration toward always placing at the top or always placing at the bottom.

11 Boundary of the universality class

The nonconcentration conditions cannot simply be removed.

For standardized permutations, take a two-letter alphabet with

$$p_{n,0} = 1 - \delta_n, \quad p_{n,1} = \delta_n,$$

where $n\delta_n \rightarrow 0$. With probability tending to one, every letter is zero, so the standardized permutation is the identity and its order is one. Thus $\|p^{(n)}\|_\infty \rightarrow 1$ can destroy the Erdős–Turán law completely.

For shelf shuffles, the endpoint parameters can also destroy the conclusion. When $m = 1$, the rules $p = 0$ and $p = 1$ produce, respectively, the identity and the reversal permutation (up to the harmless convention for reading the deck); their orders are bounded by two. Thus the theorem identifies a genuine uniform nondegeneracy regime. It does not assert universality throughout every possible boundary scaling.

A natural next question is to classify the transition when the exponential stability parameter $\rho = \rho_n$ tends to one. In the standardized model this means that one letter becomes dominant. In the shelf model it means p_n approaches zero or one. Such regimes may exhibit different order scales and are not addressed here.

12 Consequences and further directions

The results establish a reusable principle:

An exponentially localized perturbation of the Ewens cycle index may change every fixed short-cycle law while leaving the entire logarithmic Gaussian cycle field, and hence the Erdős–Turán order process, unchanged.

The theorem applies with no convergence of the underlying parameter profile. It is therefore robust under triangular arrays of biased riffle probabilities, alphabet sizes, shelf numbers, and shelf asymmetries within the stated compact regime.

The scalar order-level problem is settled through the first Edgeworth term: the sharp rate, its optimality, and the universal leading Kolmogorov constant are all identified. The long-cycle total-variation theorem also suggests quantitative finite-dimensional extensions, but a sharp multivariate or path-space Berry–Esseen theorem requires a choice of metric and a separate Gaussian approximation argument; it is not claimed here. A second direction is to weaken exponential stability to polynomial or operator-theoretic mixing of the cycle-index coefficients. That extension would be relevant to dense kernel/permanental permutations and weak Mallows permutations.

A A coefficient-tail lemma

We record the elementary estimate used in Proposition 4.1.

Lemma A.1. *Let q be fixed. Suppose*

$$A_j(u) = \sum_{r \leq U} a_{j,r} u^r, \quad \sum_r |a_{j,r}| \leq M \log n,$$

and let

$$E_j(u) = \sum_{m \geq b} e_{j,m} u^m, \quad \sum_{m \geq b} |e_{j,m}| \leq M \rho^b$$

for $1 \leq j \leq q$, where $\rho < 1$. If $qU < n$, then the difference between the sum of coefficients through degree n and the value at $u = 1$ of any product of at most q factors selected from the $A_j + E_j$ is

$$O_{q,M}((1 + \log n)^q \rho^b).$$

Proof. The product containing only A factors has degree at most $qU < n$ and therefore contributes no tail. Every other product contains an E factor. Bound its total absolute coefficient mass by the product of the ℓ^1 coefficient norms. $\square$

B A tightness criterion with vanishing errors

The following standard variant of Billingsley’s two-increment criterion is convenient for triangular arrays.

Lemma B.1. *Let X_n be cadlag processes. Suppose their finite-dimensional distributions are tight, their maximal jumps converge to zero in probability, and there are $C < \infty$ and $\eta_n \rightarrow 0$ such that*

$$\mathbb{E} \left[|X_n(t) - X_n(s)|^2 |X_n(u) - X_n(t)|^2 \right] \leq C(u - s)^2 + \eta_n$$

for all $0 \leq s \leq t \leq u \leq 1$. Then (X_n) is tight in $D[0, 1]$, and every subsequential limit is continuous.

Proof. Fix a mesh width $\delta > 0$. In the proof of Billingsley’s two-increment criterion, the interval $[0, 1]$ is first covered by a finite mesh depending only on δ , and Markov’s inequality is applied to pairs of adjacent increments. Replacing the usual bound $C(u - s)^2$ by $C(u - s)^2 + \eta_n$ adds at most $C_\delta \eta_n$ to the resulting probability estimate, where $C_\delta < \infty$ is independent of n . This additional term vanishes when $n \rightarrow \infty$ with δ fixed. The remainder of the proof of [5, Theorem 13.5] is unchanged and gives the required cadlag modulus after $\delta \downarrow 0$. The maximal-jump assumption rules out discontinuities in every subsequential limit. $\square$

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